

WASP-50b

R-1124

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_201114 · created 2026-07-30T22:18:16+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459167.7155 +/- 0.0037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.015
Transit depth (Rp/Rs)^2	3.05 +/- 0.52 [%]
Orbital Inclination (inc)	84.63 +/- 1.0
Airmass coefficient 1 (a1)	1.007 +/- 0.011
Airmass coefficient 2 (a2)	-0.005 +/- 0.015
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#2 - [278, 211]
Optimal Method	PSF photometry
Transit Duration (day)	0.0738 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.175 ± 0.015 vs 0.1404 (deviation 2.31σ)
Duration fit vs published	0.0738 d vs 0.0754 d (deviation 0.23σ)
Mid-time O–C	1.9 min
Depth SNR	11.7
Residual scatter	1.11 %

Light curve

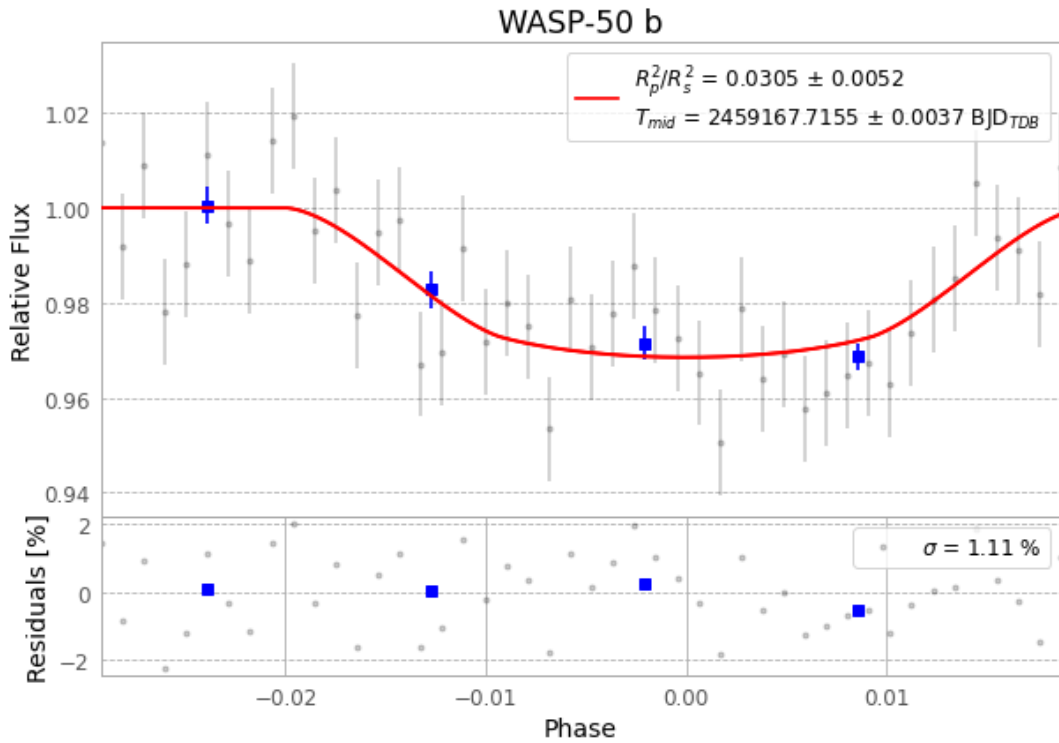


Figure 1 — FinalLightCurve_WASP-50 b_2020-11-13.png (SHA-256 cc3c1caa449e4e95...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 234], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b871de271ee2511b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

46 FITS frames (30 MB), WASP-50201114033914.FITS → WASP-50201114055412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-201114.log (3e213d079840...), FinalLightCurve_WASP-50 b_2020-11-13.png (cc3c1caa449e...), FinalLightCurve_WASP-50 b_2020-11-13.csv (a0d425161ad9...)

Compute resources

Wall time 85.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 22:18Z.